

Xudong Zhu

Ph.D. Student in Computer Science · The Ohio State University · Columbus, OH

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I study representation learning and training dynamics in large language models, and use these perspectives to advance mechanistic interpretability: how features and behaviors are encoded in representation space, and how that structure enables reliable interpretation and control.

RESEARCH INTERESTS

Representation Learning · Interpretability · Representation & Behavior Steering · Training Dynamics · Agent Behavior Control

EDUCATION

The Ohio State University

Ph.D. in Computer Science and Engineering, advised by Prof. Zhihui Zhu

2024 – Present

Columbus, OH

University of Electronic Science and Technology of China

B.S. in Computer Science · GPA 3.98 / 4.00

2020 – 2024

Chengdu, China

PUBLICATIONS

AbsTopK: Rethinking Sparse Autoencoders for Bidirectional Features

Xudong Zhu, Mohammad Mahdi Khalili, Zhihui Zhu

[ICLR 2026](#) · [Paper](#) · [Code](#)

From Emergence to Control: Probing and Modulating Self-Reflection in Language Models

Xudong Zhu, Jiachen Jiang, Mohammad Mahdi Khalili, Zhihui Zhu

[Transactions on Machine Learning Research \(TMLR\)](#) · [Paper](#) · [Code](#)

FCDS: Fusing Constituency and Dependency Syntax into Document-Level Relation Extraction

Xudong Zhu, Zhao Kang, Bei Hui

[LREC-COLING 2024](#) · [Paper](#) · [Code](#)

Alleviating Subgraph-Induced Oversmoothing in Link Prediction via Coarse Graining

Xudong Zhu, Dong Hao, Ziqin Gao, Liming Pan

[Neurocomputing, 2025](#) · [Paper](#) · [Code](#)

HONORS & RESEARCH SUPPORT

MMLS 2026 Travel Award, Midwest Machine Learning Symposium

2026

NAIRR Pilot Project NAIRR260106 (contributor), National AI Research Resource Pilot

Geometry and Training Dynamics of Representations in Large Language Models

2026

OPEN SOURCE

LLMUsage · Local-first LLM subscription usage monitor for Hyprland (OpenAI, GLM, Grok, Claude)

TypeScript

dotfiles · fish, Neovim, and OpenCode configuration managed with GNU Stow

Lua